

DLR-TomoSAR — Physics-constrained learning

five loss terms · curriculum · the unrolled solver

July 2026

Why constrain the solution space

- The labels are classical fits: the network's ceiling is the label quality — and Act IV showed the network happily leans on statistical shortcuts.
- Physics terms restrict predictions to profiles **consistent with the SAR signal model**:
 - today — as a regularizer: physical quantities recomputed from the predicted profile must match those of the GT profile;
 - next — as self-supervision against the *measured* covariance itself, removing the label ceiling entirely.
- Five terms implemented and theory-tested; every one differentiable end-to-end through the Gaussian-mixture curve reconstruction.

The shared forward operator

All terms use the vertical wavenumber and the steering matrix on the elevation grid:

$$k_z^{(i)} = \frac{4\pi b_i}{\lambda r_0 s}, \quad A_{in} = e^{j k_z^{(i)} \xi_n}$$

- Geometry resolved from campaign track parameters — not nominal defaults: per-pass perpendicular baselines, look angle, slant range.
- **Per-pixel k_z field** for range/azimuth-varying geometry, built at preprocessing and threaded dataset → trainer → loss.
- Steering sign settled empirically ($+k_z$: median peak error 0.67 m, within one Rayleigh cell).

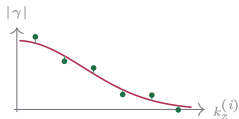
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

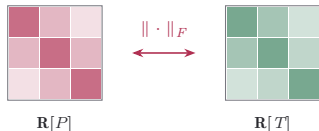
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

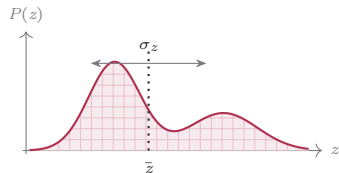
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n) \Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



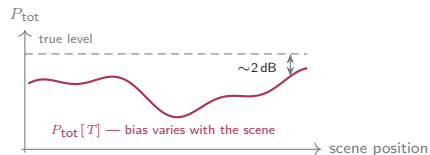
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta\xi$$

$$\ell_{\text{pow}} = \left\langle (P_{\text{tot}}[P] - P_{\text{tot}}[T])^2 \right\rangle$$

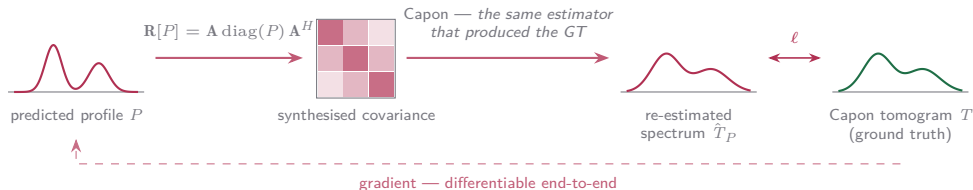


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$

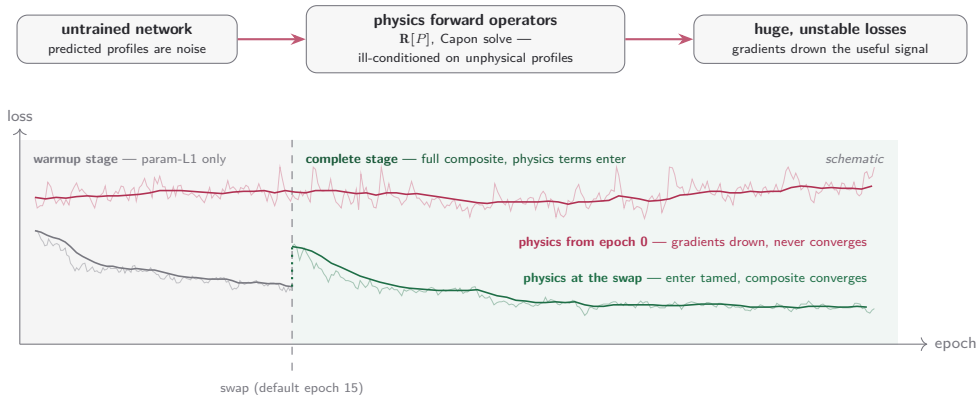


- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

Why the physics terms need a curriculum

The physics terms are the most sensitive consumers of the reconstructed profile — and at random initialisation they see only noise:



Swap mechanics, calibration — and status

Each detail is an anticipated failure mode:

- **Resets at the swap:** LR schedule origin, LR warmup, Adam moments, early-stopping counters — optimizer state calibrated to the old loss is stale under the new one.
- **Checkpoint-baseline reset:** best-loss tracking restarts across the non-comparable loss scales — otherwise the pre-swap best could never be beaten.
- **Stage inheritance:** the complete stage is the base configuration, warmup a derived pre-phase materialised at launch — the stages cannot drift.
- **Loss-scale probe:** every active term run at unit weight over probe batches, IQR-filtered, folded into the per-term weights — physics terms enter at calibrated magnitude, not guessed weights.

Status

Implemented, SAR-theory-verified, geometry wired end-to-end — the single-term $K=2$ sweep is in (next frames); the Capon-cycle arm, the most expensive of the five, is still queued.

The physics sweep at $K=2$ — every term rides low, the scale terms collapse high

objective	reconstruction		detection (matched)						
	curve $R^2 \uparrow$	cos med $\uparrow$	prec. $\uparrow$	recall $\uparrow$	F1 $\uparrow$	exact $\uparrow$	under $\downarrow$	over $\downarrow$	peak p95 $\downarrow$
param-L1 + cos	0.747 $\pm$.006	0.965 $\pm$.004	0.895 $\pm$.008	0.901 $\pm$.001	0.898 $\pm$.004	0.908 $\pm$.007	0.041 $\pm$.003	0.051 $\pm$.009	4.7 $\pm$.0
+ coherence re-synthesis									
$w=0.01$	0.752 $\pm$.004	0.966 $\pm$.002	0.895 $\pm$.006	0.902 $\pm$.001	0.898 $\pm$.003	0.910 $\pm$.002	0.040 $\pm$.004	0.050 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.752 $\pm$.006	0.966 $\pm$.001	0.893 $\pm$.005	0.902 $\pm$.001	0.898 $\pm$.002	0.911 $\pm$.003	0.039 $\pm$.003	0.051 $\pm$.004	4.7 $\pm$.0
$w=0.25$	0.755 $\pm$.007	0.966 $\pm$.002	0.892 $\pm$.009	0.903 $\pm$.002	0.897 $\pm$.004	0.909 $\pm$.005	0.038 $\pm$.005	0.053 $\pm$.010	4.7 $\pm$.0
$w=0.5$	0.748 $\pm$.009	0.965 $\pm$.003	0.886 $\pm$.013	0.903 $\pm$.001	0.895 $\pm$.007	0.911 $\pm$.010	0.032 $\pm$.006	0.057 $\pm$.014	4.7 $\pm$.0
$w=1$	0.751 $\pm$.008	0.966 $\pm$.002	0.870 $\pm$.045	0.903 $\pm$.001	0.886 $\pm$.024	0.885 $\pm$.063	0.032 $\pm$.005	0.083 $\pm$.068	4.7 $\pm$.0
+ covariance matching									
$w=0.01$	0.752 $\pm$.001	0.965 $\pm$.002	0.896 $\pm$.005	0.901 $\pm$.001	0.898 $\pm$.003	0.906 $\pm$.004	0.043 $\pm$.001	0.051 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.748 $\pm$.001	0.966 $\pm$.002	0.899 $\pm$.005	0.902 $\pm$.001	0.900 $\pm$.002	0.909 $\pm$.004	0.043 $\pm$.002	0.048 $\pm$.005	4.7 $\pm$.0
$w=0.25$	0.738 $\pm$.004	0.966 $\pm$.002	0.896 $\pm$.006	0.899 $\pm$.002	0.898 $\pm$.003	0.902 $\pm$.008	0.045 $\pm$.002	0.053 $\pm$.008	4.7 $\pm$.0
$w=0.5$	0.727 $\pm$.011	0.965 $\pm$.002	0.900 $\pm$.001	0.898 $\pm$.002	0.899 $\pm$.001	0.907 $\pm$.003	0.046 $\pm$.003	0.047 $\pm$.001	4.7 $\pm$.0
$w=1$	0.723 $\pm$.006	0.965 $\pm$.001	0.899 $\pm$.005	0.899 $\pm$.002	0.899 $\pm$.002	0.903 $\pm$.003	0.047 $\pm$.004	0.050 $\pm$.006	4.7 $\pm$.0
+ moments									
$w=0.01$	0.753 $\pm$.003	0.966 $\pm$.002	0.895 $\pm$.006	0.902 $\pm$.001	0.898 $\pm$.003	0.910 $\pm$.004	0.040 $\pm$.005	0.050 $\pm$.006	4.7 $\pm$.0
$w=0.05$	0.755 $\pm$.002	0.966 $\pm$.002	0.896 $\pm$.005	0.902 $\pm$.001	0.899 $\pm$.002	0.910 $\pm$.004	0.041 $\pm$.004	0.049 $\pm$.005	4.7 $\pm$.0
$w=0.25$	0.751 $\pm$.002	0.965 $\pm$.002	0.897 $\pm$.004	0.901 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.003	0.044 $\pm$.000	0.049 $\pm$.004	4.7 $\pm$.0
$w=0.5$	0.750 $\pm$.004	0.966 $\pm$.002	0.898 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.001	0.909 $\pm$.001	0.044 $\pm$.002	0.047 $\pm$.003	4.7 $\pm$.0
$w=1$	0.746 $\pm$.005	0.966 $\pm$.001	0.897 $\pm$.003	0.901 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.003	0.043 $\pm$.001	0.049 $\pm$.003	4.7 $\pm$.0
+ total power									
$w=0.01$	0.752 $\pm$.003	0.965 $\pm$.002	0.896 $\pm$.005	0.902 $\pm$.001	0.899 $\pm$.002	0.910 $\pm$.004	0.041 $\pm$.004	0.049 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.751 $\pm$.005	0.966 $\pm$.001	0.897 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.002	0.044 $\pm$.001	0.048 $\pm$.002	4.7 $\pm$.0
$w=0.25$	0.747 $\pm$.003	0.965 $\pm$.001	0.898 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.001	0.908 $\pm$.004	0.044 $\pm$.003	0.049 $\pm$.004	4.7 $\pm$.0
$w=0.5$	0.737 $\pm$.015	0.965 $\pm$.002	0.900 $\pm$.005	0.899 $\pm$.002	0.899 $\pm$.003	0.905 $\pm$.006	0.048 $\pm$.004	0.047 $\pm$.006	4.7 $\pm$.0
$w=1$	0.724 $\pm$.020	0.963 $\pm$.005	0.902 $\pm$.003	0.898 $\pm$.005	0.900 $\pm$.001	0.907 $\pm$.003	0.049 $\pm$.007	0.044 $\pm$.004	4.8 $\pm$.3

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = loss (param-L1 + 0.5 cos + w phys) / (1.5 + w), one physics term per arm; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. Val losses do not compare across arms (different composites); curve MAE / PSNR / SSIM stay flat outside the collapse arms (next frames); peak p95 is pinned at 4.7. **bold** = best in column, marked only where best and worst differ by more than 5%. **Green**: coherence $w=0.25$ takes the best curve R^2 ; at $w=1$ its gate trades misses for false alarms (under 0.041 $\rightarrow$ 0.032, over 0.051 $\rightarrow$ 0.083; seed lottery: precision $\pm$.045, exact $\pm$.063). **Red** = absolute-scale collapse past $w=0.25$ (covariance, total power). cos med: median per-pixel cosine to GT: peak p95: 95th pct of the per-pixel peak error.

The $K=2$ physics by scatterer count — the scatterer set is untouched

metric	physics term at its weight					
	none	coh. $w=0.25$	coh. $w=1$	mom. $w=0.05$	cov. $w=0.05$	pow. $w=0.05$
precision $\uparrow$	0.895 $\pm$.008	0.892 $\pm$.009	0.870 $\pm$.045	0.896 $\pm$.005	0.899 $\pm$.005	0.897 $\pm$.003
$k=1$	0.935 $\pm$.010	0.932 $\pm$.011	0.902 $\pm$.067	0.937 $\pm$.006	0.938 $\pm$.006	0.938 $\pm$.003
$k=2$	0.828 $\pm$.006	0.826 $\pm$.006	0.818 $\pm$.009	0.828 $\pm$.006	0.834 $\pm$.003	0.831 $\pm$.004
recall $\uparrow$	0.901 $\pm$.001	0.903 $\pm$.002	0.903 $\pm$.001	0.902 $\pm$.001	0.902 $\pm$.001	0.900 $\pm$.001
$k=1$	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000
$k=2$	0.763 $\pm$.002	0.766 $\pm$.004	0.768 $\pm$.001	0.764 $\pm$.003	0.763 $\pm$.003	0.761 $\pm$.003
<i>count acc pred k</i>						
pred $k=1 \uparrow$	0.944 $\pm$.004	0.948 $\pm$.006	0.954 $\pm$.002	0.944 $\pm$.005	0.943 $\pm$.002	0.941 $\pm$.001
pred $k=2 \uparrow$	0.814 $\pm$.024	0.808 $\pm$.023	0.755 $\pm$.122	0.817 $\pm$.015	0.819 $\pm$.015	0.819 $\pm$.007
μ MAE $\downarrow$	1.71 $\pm$.07	1.72 $\pm$.06	1.81 $\pm$.12	1.69 $\pm$.04	1.65 $\pm$.02	1.67 $\pm$.03
$k=1$	0.36 $\pm$.01	0.35 $\pm$.00	0.35 $\pm$.00	0.35 $\pm$.01	0.35 $\pm$.01	0.35 $\pm$.00
$k=2$	3.78 $\pm$.14	3.82 $\pm$.13	4.02 $\pm$.26	3.75 $\pm$.09	3.66 $\pm$.05	3.72 $\pm$.08
σ MAE $\downarrow$	0.774 $\pm$.011	0.778 $\pm$.011	0.809 $\pm$.020	0.771 $\pm$.018	0.754 $\pm$.007	0.761 $\pm$.005
$k=1$	0.193 $\pm$.002	0.192 $\pm$.003	0.194 $\pm$.006	0.195 $\pm$.003	0.194 $\pm$.002	0.194 $\pm$.003
$k=2$	1.668 $\pm$.023	1.675 $\pm$.016	1.737 $\pm$.035	1.657 $\pm$.043	1.620 $\pm$.017	1.639 $\pm$.015
a MAE $\downarrow$	0.182 $\pm$.004	0.182 $\pm$.005	0.185 $\pm$.006	0.182 $\pm$.003	0.187 $\pm$.005	0.184 $\pm$.004
$k=1$	0.058 $\pm$.003	0.058 $\pm$.004	0.061 $\pm$.008	0.058 $\pm$.003	0.061 $\pm$.004	0.059 $\pm$.004
$k=2$	0.373 $\pm$.006	0.373 $\pm$.005	0.373 $\pm$.004	0.372 $\pm$.002	0.381 $\pm$.007	0.377 $\pm$.004

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = physics term at its best-scoring weight, plus the $w=1$ coherence arm; none = the no-physics reference. All rows matched (Hungarian pairing); cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only. **bold** = best in row, marked only where best and worst differ by more than 5% — almost nothing clears it: detection, count calibration and component errors all sit inside combined seed noise, so the physics terms shape the profile, not the scatterer set. The one exception is full-weight coherence: its wider gate pays μ/σ (μ 1.71 $\rightarrow$ 1.81, σ 0.774 $\rightarrow$ 0.809) and destabilises the claimed pairs (pred $k=2$ acc $\pm$.122). Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

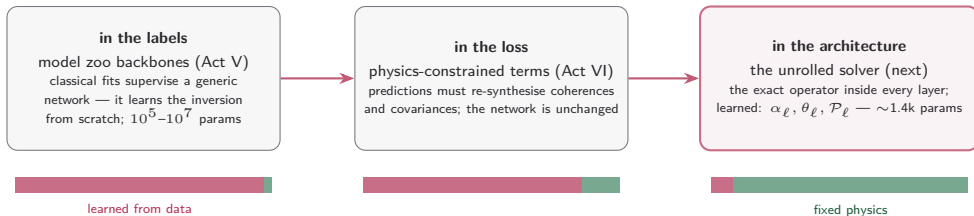
The $K=2$ physics per-pixel tail — descriptor and covariance terms cap the blow-ups

metric	physics term at its weight					
	none	coh. $w=0.25$	coh. $w=1$	mom. $w=0.05$	cov. $w=0.05$	pow. $w=0.05$
<i>curve level</i>						
curve R^2 $\uparrow$	0.747 $\pm$.006	0.755 $\pm$.007	0.751 $\pm$.008	0.755 $\pm$.002	0.748 $\pm$.001	0.751 $\pm$.005
curve MAE $\downarrow$	0.0370 $\pm$.0005	0.0366 $\pm$.0003	0.0371 $\pm$.0007	0.0365 $\pm$.0002	0.0365 $\pm$.0002	0.0366 $\pm$.0001
RMSE $\downarrow$	0.217 $\pm$.002	0.213 $\pm$.003	0.215 $\pm$.003	0.214 $\pm$.001	0.217 $\pm$.000	0.215 $\pm$.002
PSNR [dB] $\uparrow$	52.63 $\pm$.10	52.78 $\pm$.13	52.70 $\pm$.14	52.76 $\pm$.04	52.65 $\pm$.02	52.70 $\pm$.08
<i>SSIM vs GT</i>						
elev $\uparrow$	0.9671 $\pm$.0004	0.9674 $\pm$.0002	0.9670 $\pm$.0006	0.9674 $\pm$.0002	0.9675 $\pm$.0002	0.9674 $\pm$.0001
range $\uparrow$	0.9510 $\pm$.0008	0.9516 $\pm$.0004	0.9499 $\pm$.0027	0.9519 $\pm$.0004	0.9521 $\pm$.0004	0.9519 $\pm$.0004
azim $\uparrow$	0.9610 $\pm$.0006	0.9615 $\pm$.0003	0.9606 $\pm$.0014	0.9615 $\pm$.0003	0.9616 $\pm$.0002	0.9615 $\pm$.0003
<i>per-pixel R^2 — the tail</i>						
median $\uparrow$	0.888 $\pm$.007	0.891 $\pm$.006	0.870 $\pm$.049	0.888 $\pm$.009	0.889 $\pm$.009	0.889 $\pm$.007
mean $\uparrow$	-1.17 $\pm$.79	-0.88 $\pm$.53	-3.71 $\pm$ 4.34	+0.53 $\pm$.19	+0.77 $\pm$.01	+0.73 $\pm$.05
p_5 $\uparrow$	0.275 $\pm$.065	0.284 $\pm$.021	-0.912 $\pm$ 2.630	0.355 $\pm$.008	0.394 $\pm$.003	0.377 $\pm$.012
p_1 $\uparrow$	-14.88 $\pm$ 11.38	-12.37 $\pm$ 4.01	-50.17 $\pm$ 70.98	-2.16 $\pm$ 1.08	-0.45 $\pm$.06	-0.87 $\pm$.35
min $\uparrow$	-13.2k $\pm$ 3.3k	-12.2k $\pm$ 4.8k	-22.5k $\pm$ 11.5k	-4.4k $\pm$ 1.2k	-4.2k $\pm$ 0.2k	-4.1k $\pm$ 0.1k
<i>per-pixel cosine</i>						
median $\uparrow$	0.965 $\pm$.004	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.001
mean $\uparrow$	0.934 $\pm$.003	0.936 $\pm$.001	0.936 $\pm$.001	0.935 $\pm$.001	0.934 $\pm$.001	0.934 $\pm$.001
p_1 $\uparrow$	0.588 $\pm$.011	0.599 $\pm$.009	0.601 $\pm$.007	0.584 $\pm$.006	0.574 $\pm$.007	0.579 $\pm$.008

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = physics term at its best-scoring weight, plus the $w=1$ coherence arm; none = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px; pixel R^2 min and its spread in thousands (k). **bold** = best in row, marked only where best and worst differ by more than 5%. The typical pixel never moves (pixel R^2 median 0.888–0.891, cos med 0.965–0.966, SSIM in the fourth decimal on all three axes) — the story is the tail: a handful of catastrophic pixels wreck the no-physics mean (-1.17, p_1 -14.9, min -13.2k), and the descriptor and covariance terms cap the blow-ups (covariance $w=0.05$: mean +0.77, p_1 -0.45, min -4.2k) while cutting the curve-metric seed spread up to $\sim 5 \times$. Coherence is NOT tail insurance — at $w=1$ its tail is worse than the baseline (mean -3.71, min -22.5k): the Fourier-domain term shapes coherences, not pixel-domain outliers. The curve-level and SSIM blocks carry the headline claim: the gains that exist sit at the edge of combined seed noise.

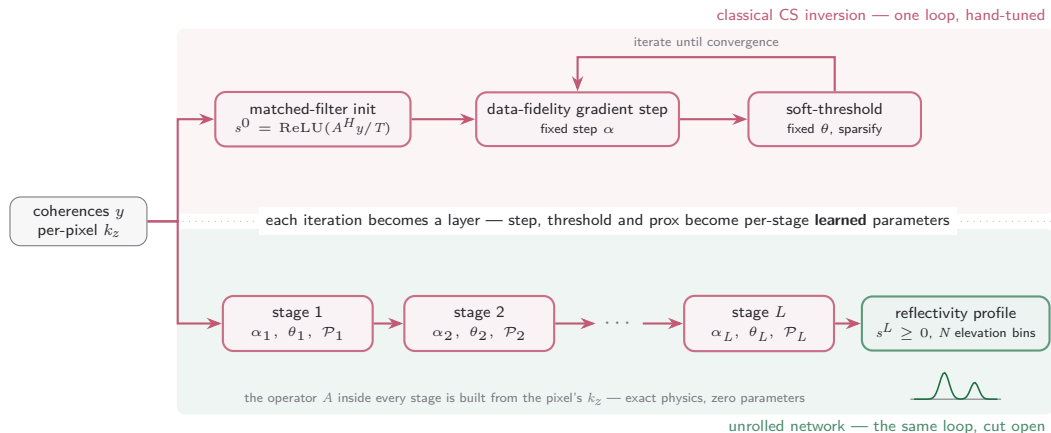
Context: where does the physics live?

Three places the SAR signal model can enter a learned inversion — the project now covers all three:



The unrolled family probes the far end of the axis — **the question it answers:** how much of the zoo's capacity is spent re-learning physics we already know?

The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \operatorname{Re}[A^H(y - As^\ell)]/L_{\text{lip}}, \quad s^{\ell+1} = \operatorname{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

γ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\text{lip}} = TN \mathrm{d}z^2$ normalises the step. Default $L=8$: $\sim 1.4\text{k}$ parameters.

Adapting γ -Net — a different unknown, a per-pixel operator

γ -Net — one scene geometry, weights learned



a fixed matrix can be learned — an operator that changes at every pixel cannot



this project — per-pixel geometry, physics exact, scalars learned

γ -Net inverts $g = \mathbf{R}\gamma + \varepsilon, \quad \gamma \in \mathbb{C}^N$

this project inverts $y = A(x)s, \quad s \in \mathbb{R}_{\geq 0}^N$

g : one complex look per track (first-order); y : per-track coherences (second-order) — averaging interferograms cancels the speckle phase, so the recoverable object is the power profile s , not the complex reflectivity γ . ε : measurement noise; T : tracks; N : elevation bins; $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation.

Adapting γ -Net — mechanism by mechanism

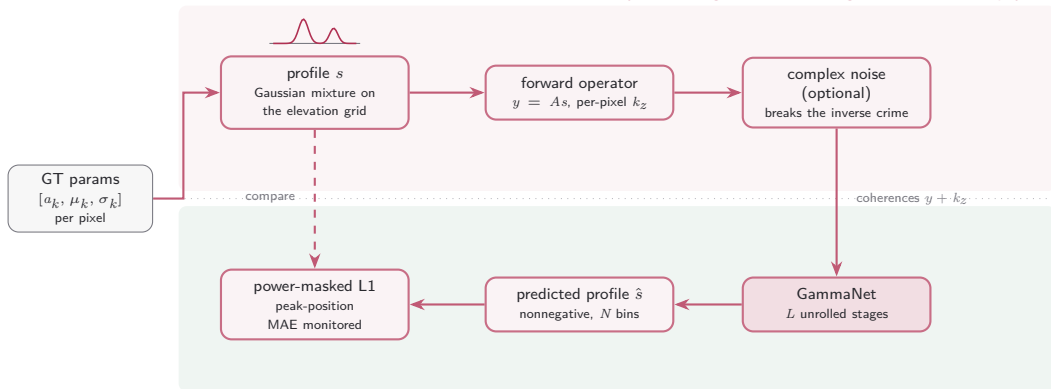
mechanism	γ -Net (Qian et al. 2022)	this project	why
iteration	gradient step $\rightarrow$ shrinkage, L stages	= kept	the unrolling premise itself
supervision	train on simulated measurements	= kept	the exact operator generates the pairs
unknown	complex reflectivity $\gamma \in \mathbb{C}^N$	$\rightarrow$ power profile $s \in \mathbb{R}_{\geq 0}^N$	speckle phase averages out of coherences
measurements	single-look stack g	$\rightarrow$ per-track coherences y	the pipeline is second-order end to end
operator	one fixed $\mathbf{R}$, scene-wide	$\rightarrow$ $A(x)$ from the pixel's k_z	airborne geometry varies across the scene
learned weights	$\mathbf{W}^\ell$ per stage, coupled to $\mathbf{R}$	$\rightarrow$ none — scalars α_ℓ, θ_ℓ	a per-pixel matrix cannot be learned
shrinkage	support selection + piecewise-linear function	$\rightarrow$ prox $\mathcal{P}_\ell$, then $\text{ReLU}(\cdot - \theta_\ell)$	the prior must respect $s \geq 0$
model order	Bayesian information criterion after inversion	$\rightarrow$ downstream parameter extraction	the pipeline already fits K scatterers

The contract every change preserves: **the physics stays exact inside the network — only the prior is learned.**

support selection: the largest-magnitude entries bypass the threshold at each stage; piecewise-linear function: γ -Net's learned replacement for soft-thresholding; coupled weights: $\mathbf{W}_2^\ell = \mathbf{I} - \mathbf{W}_1^\ell \mathbf{R}$ halves the free parameters per stage. γ -Net reaches its accuracy at 12 stages; default here $L = 8$. **green** = exact physics carried by the operator · **red** = learned · grey = kept from γ -Net.

Training the unrolled model against the exact operator

synthesis — ground truth through the exact forward physics



inversion — the unrolled network runs it backwards

Phase 1 (implemented): learn to invert the *exact* operator on synthesised coherences. Phase 2 (pending): measured covariances from the stack — no label ceiling.

Isolated family — own registry, trainer and entry point (`train_unrolled`); optimizer splits steps/thresholds from the prox; no AMP, the operator is complex-valued.

Why it exists: the physics-native comparison point — if $\sim 1.4k$ parameters of encoded physics match capacity-matched vision backbones, those backbones are mostly re-learning known physics.